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MILITARY SPECIFICATION

COMPUTER, GUN DIRECTION: M18; MANUFACTURING PROCESS AND INSPECTION OF

1. SCOPE

1.1 This specification covers requirements of the manufacturing processes and inspection for construction of Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-T-236	Tetrachloroethylene (Perchloroethylene), Technical Grade
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
TT-N-95	Naptha, Aliphatic
TT-S-735	Standard Test Fluids, Hydrocarbon

Military

MIL-T-27	Transformers and Inductors (Audio, Power, and High Power Pulse) General Specification for
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-T-713	Twine, Impregnated, Lacing and Tying (For use in Electrical and Electronic Equipment)
MIL-T-3530	Treatments, Mildew Resistant and Water Repellent for Thread and Twine
MIL-L-6806	Lacquer, Clear, Aluminum Clad Aluminum Alloy Surfaces

FSC 1220

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MIL-S-6872	Soldering Process, General Specification for
MIL-V-6894	Varnish, Oil Type, Gloss Finish, Glyceryl Phthalate Base
MIL-T-7003	Trichloroethylene, Stabilized Degreasing
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-13949	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring)
MIL-F-14256	Flux, Soldering Liquid (Rosin Base)
MIL-P-18177	Plastic Sheet, Laminated, Thermosetting, Glass Fiber Base, Epoxy-Resin
MIL-P-19833	Plastic Molding Material and Plastic Molded Parts, Glass Fiber Filled, Diallylphthalate Resin
MIL-I-23053	Insulation Sleeving, Electrical, Flexible, Heat Shrinkable
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-G-45204	Gold Plating (Electrodeposited)
MIL-I-45607	Inspection Equipment, Supply and Maintenance, for Ordnance

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-681	Identification Coding and Application of Hookup and Lead Wire

DRAWING

U.S. Army Munitions Command

F10526101	Computer, Gun Direction: M18

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials

ASTM D523-53T	Method of Test for Specular Gloss
ASTM D750-55T	Method of Test for Resistance to Accelerated Light Aging of Rubber Compounds

(Applications for copies should be addressed to American Society for Testing Materials, 1916 Race St., Phila., Pa. 19103.)

3. REQUIREMENTS

3.1 Order of precedence. - Order of precedence shall be as established in the detail specification with the following exception; should any conflict exist between the detail specification and this specification, the detail specification shall take precedence.

3.2 Etched circuit assembly.

3.2.1 Circuit board materials. - The circuit board materials shall be as follows:

(a) Laminated reinforced plastic sheet whose properties conform to MIL- P-18177.

(b) Electrolytic copper shall be 2 oz per square foot (foil thickness of $0.0028 \begin{smallmatrix} + 0.0007 \\ - 0.0003 \end{smallmatrix}$ inch) with minimum purity of 99.5 percent.

(c) Copper clad laminate sheets shall be $0.062 \pm .003$ inch.

(d) The copper clad material shall be free from blisters, wrinkles, cracks, holes, dents, scratches and other defects that will affect appearance and serviceability. Unless otherwise specified the copper clad plastic sheet covered by this specification shall conform to MIL-P-13949, Type FL-GE-0.062-C-2/2 Cl-III or FL-GE-0.062-C-2/0 Cl-III as required by applicable drawings.

3.2.2 Installation of hardware. - All installation of hardware shall be performed by swaging, unless other methods of installation are specifically called out on the drawing. The assembly shall not contain any of the following defects:

(a) Cracks or deformation of the circuit board adjacent to installed hardware.

(b) Cracks, ruptures, or delamination of the circuit board.

(c) Bent terminals or mechanical standoffs.

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(d) Loose or improperly installed hardware (swaged type).

(e) Damage to the rollover flange of the swaged hardware such that two or more full-length cracks are located within a 45 degree angle.

NOTE: Cracks which do not extend more than halfway through the rollover flange (from the periphery to the inside of the rollover) shall be disregarded.

(f) Terminals and standoffs whose rollover flange is not concentric to or is not 30 percent larger than the terminal.

(g) Hardware from which the plating has been removed or peeled.

(h) Damage or permanently indented collars of press type terminals.

(i) Curvature along the top and bottom edges of the circuit board exceeding 0.0321 inch or the twist warpage exceeding 0.020 inch on the connector side.

(j) Radius of inside corners or cutouts more than 5/32 inch or less than 3/32 inch.

3.2.3 Plating and etching.

3.2.3.1 Cleaning.- The board shall be clean and free from dirt, oil corrosion products, salts, and other foreign matter. Holes used as plated-through connections shall be free of foreign matter.

3.2.3.2 Plating.- The copper covered laminate shall be plated with one of the following:

(a) Gold plating.- There shall be 0.00007 inches (70 microinches) minimum gold on all copper surfaces, if gold is deposited directly on the copper. Purity of the gold deposit shall be 99.8 percent minimum. Unless otherwise specified, the gold plating shall conform to MIL-G-45204, Type I.

(b) Gold-nickel plating.- There shall be 0.00005 inch (50 microinches) minimum gold over 0.0002 inch minimum nickel if gold and nickel are deposited over the copper. Unless otherwise specified, the nickel plating shall conform to QQ-N-290, Class 2. The gold plating shall be as specified in 3.2.3.2(a).

(c) Board contacts in the 0.312 engagement area shall not exceed 0.0657 inch after gold or gold-nickel plating.

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3.2.3.3 Feed-through holes.

3.2.3.3.1 Double sided boards.- The walls of all holes used as feed-through holes of double sided boards (MIL-P-13949, Type FL-GE-01062-C-2/2 CL-III), or for insertion of component leads therein shall be plated-through. There shall be a minimum of 0.00075 inch of copper in plated-through holes. The walls of the plated-through holes shall be completely plated with uniform and continuous copper and gold or copper, gold and nickel platings without voids or breaks in either the copper, gold or nickel plating. There shall be no foreign matter in plated-through holes that may resist the wetting and capillary acti n of solder. There shall be no separation visible between the surface copper foil and the copper of plated-through holes when examined under a 30 power microscope. The outer periphery of a drilled and plated-through hole shall not fall outside its islands, but may be as close as 0.005 inch to one edge.

3.2.3.3.2 Single sided boards.- The walls of all component holes on single sided boards (MIL-P-13949, Type FL-GE-0.062-C-2/0CL-III), shall not be plated-through.

3.2.3.4 Conductor pattern.- The conductor pattern shall be free of loosely adhering feather edges of metal. The finished line width shall not be more than 0.010 inch narrower than that specified on the printed circuit master. In no case shall the conductor line width be less than 0.020 inches.

3.2.3.5 Plated etched circuit boards.- The assembly shall not contain any of the following defects when viewed at 10 times magnification:

(a) Any combination of imperfections affecting the width and thickness of any conductor which reduces the cross-sectional area at that point by more than 25 percent or which reduces contact surface areas by more than 10 percent.

(b) Any separation of the conductor from the base laminate.

(c) Blisters in the conductor pattern.

(d) Delamination of the base laminate.

(e) Wrinkles in the conductor pattern.

(f) Metal in areas designated as metal-free on the master drawing.

(g) Copper or nickel visible through the gold plating.

3.2.3.5.1 Registration.- Misregistration of opposing printed circuits shall not exceed 0.010 inch.

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3.2.4 Mounting of components.- Pursuant to retaining the reliability of processed parts, all assembly techniques and processing operations required by this specification such as; cleaning, rinsing, drying, and marking shall not degrade components and product boards. Excessive force shall not be used when installing components in holes or clips. Components with visibly damaged or dented cases shall not be used.

3.2.4.1 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that reference symbols on the board are not obscured, and that the proper identification markings of each part are visible.

3.2.4.2 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with the component in contact with the board at least at one point. The component shall not be raised more than 0.031 inches from the board at any point on its contacting surface.

3.2.4.2.1 Pressure points.- There shall be no pressure points between insulated parts which depend on non-rigid insulation to separate conducting materials. Parts insulated with rigid materials shall not be mounted in such a manner that pressure points exist adjacent to sharp edges of non-insulated hardware. Pressure points between rigidly insulated parts are unacceptable. Line pressure (at least two areas of contact) between parts is acceptable.

3.2.4.2.2 Uninsulated parts.- All uninsulated parts, not electrically connected, shall be spaced as shown on the drawing; but in no case shall this spacing be less than an absolute minimum of 1/32 inch, including wire and solder build-up.

EXCEPTION: Intentional physical contact, such as between parts and their holding clamps, shall be allowed.

3.2.4.3 Parts with welded leads.- Extreme care shall be exercised in handling and installing any part having welded leads, such as tantalum capacitors. Such parts shall not have their connecting leads subjected to twisting, bending, or undue stressing at that portion of the lead between the welded joint and the body of the part. Parts showing evidence of undue stress shall not be used.

3.2.4.4 Large or heavy parts.- Large parts (over 1/2 inch diameter) shall be securely fastened to the board by mechanical means. Parts weighing 9 grams (1/3 ounce) or more shall be mechanically held in place on the chassis or board. Where holding devices consume excessive space, cementing is permissible.

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3.2.4.5 Mounting supports.- If the part is mounted on a support which is separate from that for the connecting terminals, sufficient slack shall be left in the leads to assure that the leads will not be broken by the motion of the respective supports.

3.2.4.6 Metal cased components.- Suitable insulation must be used to minimize the danger of accidental electrical contact between metal cased components and adjacent conductors.

3.2.4.7 Transistors.- Transistors are to be mounted perpendicular to the board and shall not tilt more than ten degrees from the perpendicular.

3.2.5 Lead bends.- No process of straightening, cutting, bending, inserting, or clinching of wire leads shall result in damage, change of rating or change of value of the component. Tools used to hold, straighten, or bend the leads of components shall have smooth jaws with rounded edges.

3.2.5.1 Lead length.- The minimum uniform lead length between the component body and the lead bend shall be 0.062 inch. For a self-supported part, the sum of the two lead lengths shall not exceed 1 inch. The minimum lead length for all components is dependent on the diameter of the wire. Minimum lead length for wire less than 0.029 inch diameter shall be 1/8 inch. Minimum lead length for wire 0.029 inch or greater in diameter shall be 3/16 inch. The lead length shall be measured from the junction of lead and the part (the point where the diameter of the lead increases) to the point of soldering at the terminal; or, in the case of components with welded leads, the measurement shall be made from the weld to the point of solder.

3.2.5.2 Bend radius.- The minimum bend radius for all formed leads shall be one lead diameter; the maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.2.5.3 Clinching of leads.- All wire leads which are inserted into printed circuit boards shall be clinched on the side opposite the point of insertion, in the direction of the conductor pattern where feasible. The length of the clinched end shall be such that no part of the lead shall lie beyond the terminal pad. In no case shall a clinched wire end be closer to a non-connected circuit path than the minimum spacing maintained between non-connected circuit paths on the board. The clinch shall be such that it holds the component part securely to the board so as to permit a good solderable contact.

3.2.5.4 Transistor and diode leads.- All transistor leads shall extend a minimum of 3/32 inch before any bending occurs. There shall be a minimum of 3/16 inch of lead from the transistor base to any soldered point.

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EXCEPTION: Leads of transistors mounted with a spacer, directly into a printed circuit board, shall have 1/16 inch of lead from the transistor base to any soldered point. Transistors with lead spacing of less than 0.100 inch may have bends in lead less than 3/32 inch from seal, when bend is necessary to conform to 0.100 grid. Bends shall not be made at junction of seal.

Transistor and metal cased diode leads shall be covered with sleeving and color coded per MIL-STD-681 so that not more than 1/32 inch bare lead is exposed.

EXCEPTION: Transistors mounted with a spacer require no sleeving on the leads.

3.2.5.5 Securing leads.- All leads to any one terminal shall be mechanically secure before soldering, except in the case of small terminals where mechanically secure before soldering, except in the case of small terminals where mechanically secure connections are impracticable. (For example, where a relatively large lead goes through a thin section terminal, such as on a tube socket or connector, crimping is considered impracticable). Each lead shall be direct and securely wrapped about the terminal for 3/4 to 1 turn. Overlapping upon each other about a terminal shall be avoided. Only the part of the terminal intended for wire connecting shall be used for leads.

3.2.6 Soldering and tinning.

3.2.6.1 Soldering.- Soldering shall be so executed that both a positive electrical and a strong mechanical connection is assured. All parts to be soldered shall be secured so that motion does not preclude obtaining a good solder joint. The temperature of the molten solder shall be maintained such that complete fusion of the solder alloy to the metal to be soldered shall result. The time of exposure to soldering shall be controlled to prevent damage to the components being soldered. The solder must wet all metallic surfaces with which it is in contact. Any cold joints or those showing a lack of wetting are unacceptable. On assembled printed circuit boards, solder must flow through the hole, wet the conductoron the other side and form a uniform concave fillet between the conductor and the lead wire. When flow or dip soldering, the conductor lines on the lower surface of the board must be covered with a thin, uniform, continuous film of solder. The surface of the solder must be bright with no evidence of discontinuities. However, pin holes or small spots of plating showing on conductors will be acceptable at completion of soldering, providing all solderable areas within a 0.100 inch radius are properly soldered and the amount of unsoldered area does not exceed five percent of the total conductor area. Hand soldering of printed circu t boards shall be prohibited except for terminals and repair. The solder

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joint must be clean and free from any trace of flux residues or foreign materials. The assembly shall be free of excessive solder globules, bridges, webs, and defective solder joints. The solder must not extend beyond the island on the upper surface of the board and must not ball up or extend beyond the lines on the lower surfaces. Also, the solder must not extend farther than the bend radius of the component lead. Solder shall not be permitted where such presence will interfere with the requirements for the assembly. Soldering directly to metal casings on components and parts is prohibited. Solder repairs shall not be made on printed assemblies by soldering a new component to an existing lead remaining from a previous component. All joints not in conformance with the requirements of this specification shall be disassembled, cleaned, tinned and resoldered. Unless otherwise specified, the soldering process shall conform to MIL-S-6872 using solder per QQ-S-571, Type SN60 or SN63.

3.2.6.2 Tinning.- The leads of all components, except transistors and tantalum capacitors, shall be tinned to a distance of 1/16 inch from the point of egress of the lead from the body of the component. The distance from the point of egress of the leads from the body of a transistor to the tinned surface of the leads shall be greater than 3/8 inch and less than 1/2 inch. Tantalum capacitor leads may be tinned up to the welded joint. The tinning solder shall not come into contact with that part of the lead extending from the body of the capacitor to the welded joing.

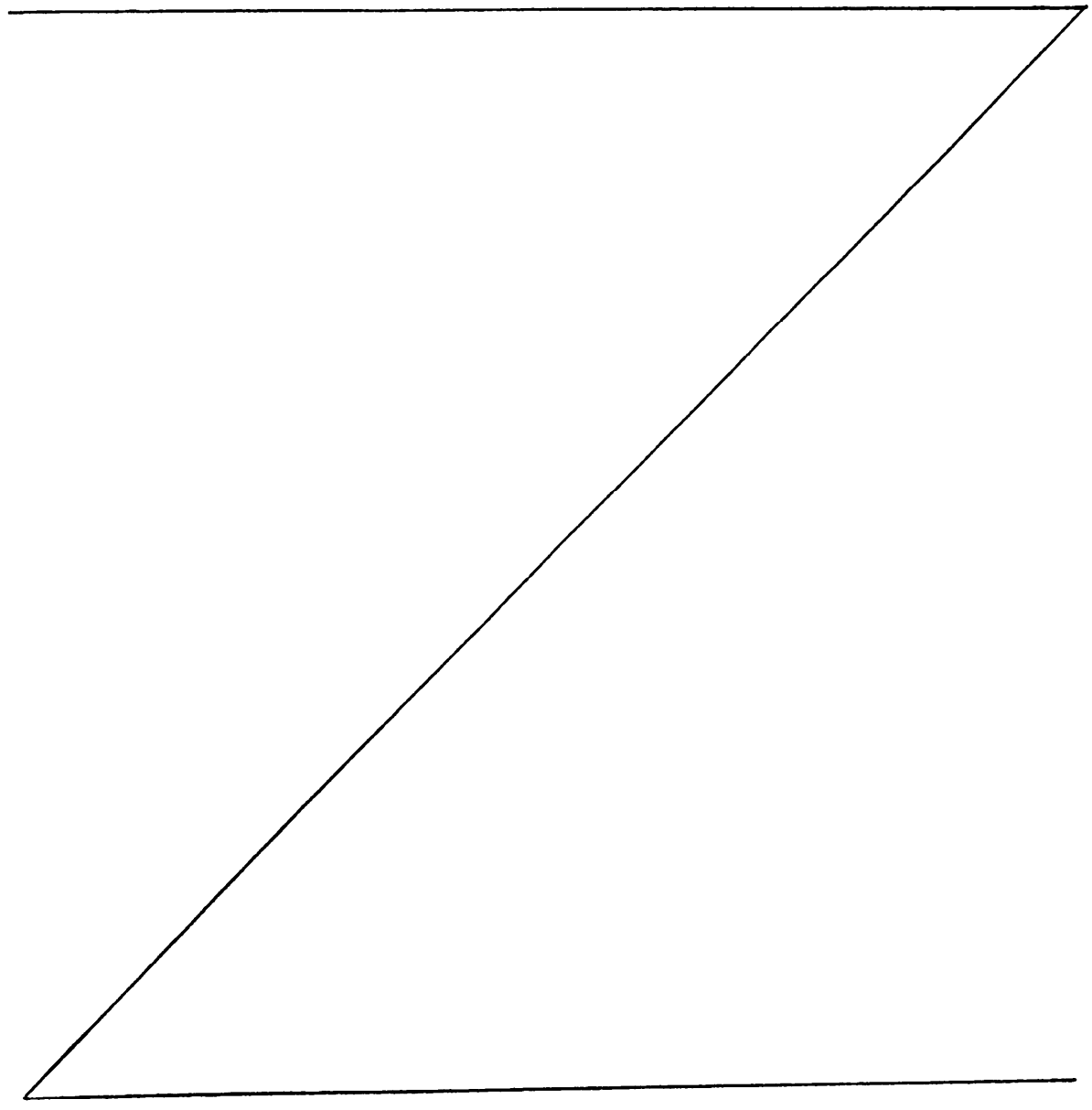
3.2.6.3 Flux.- Flux for solder applications shall be applied in a manner that will completely wet the areas to be soldered. Only the bottom side of the assembly, lead ends, and plated-through connections shall be fluxed. The fluxing technique shall be such as to prevent inadvertent deposition of flux on the component side of the assemblies. Suitable maskant shall be applied to the connector contact strip on both sides of the board before the fluxing and solder operations. The edge of the maskant shall be applied along a line formed by prolonging the edges of the end notches across the board to within 1/64 inch of the line. Unless otherwise specified, the flux shall conform to MIL-F-14256, except the flux for flow or dip soldering shall be Superior 30 or equal.

3.2.6.4 Terminals.- When terminals are used to pick up circuitry, electrical continuity shall be assured by soldering the terminal to the circuit line at the point of contact. When more than one wire or lead is soldered to a terminal post, soldering shall start from the base of the terminal and work out to the end of the terminal. Each wire or lead shall be soldered point to point followed by 5 seconds of cooling before proceeding. The solder shall be allowed to flow evenly into and around the joint forming a solder build-up about one-third the thickness of the wire. The contour of the wire shall be visible and shall not contain sharp points, globules or heavy fillets.

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3.2.6.5 Plated-through connections.- Plated-through connections of all types shall be filled with solder in order to eliminate dependence on the plating itself for circuit continuity.

3.2.6.6 Component leads.- There shall be visible evidence of component leads passing completely through the circuit board. They shall not protrude more than 0.10 inch including solder beyond the side of the circuit board opposite the component, except power transistor leads may be per manufacturing lengths.



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3.2.6.7 Heat shunts.- The use of heat shunts or cooling devices shall be mandatory when hand soldering heat sensitive components.

3.2.7 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The temperature of the cleaning solution shall not exceed 125°F. After cleaning there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.2.7.1 Drying.- The assembly shall be dried in a circulating air oven or with forced warm air. The temperature shall not exceed 165°F. Where germanium semiconductors are used, the temperature shall not exceed 125°F. Compressed air or nitrogen used for drying parts shall be filtered to meet the following requirements:

(a) Permissible dust particle size not to exceed 6 microns

(b) Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)

(c) Dust count shall not exceed 1,000 particles per cubic foot

3.2.8 Cementing and coating.

3.2.8.1 Filleting or cementing components.- If filleting or cementing of components is required, it shall be called out on the applicable drawing. Prior to filleting or cementing, the board and components shall be cleaned and dried. The filleting or cementing compound, an epoxy, shall be applied under the parts. The assembly shall then be cured and cleaned prior to applying the polyurethane coating. The cured fillet shall not exceed 3/32 inch in height. For exceptions to filleting or cementing see 3.2.8.4.

3.2.8.2 Coating.- The completed printed wiring assembly shall be covered with a non-corrosive, moisture and fungus resistant coating. The coating used shall be tough, flexible polyurethane resin. It shall provide electrical insulation, abrasion resistance, adhesion to metals and nonmetals, and protection against environmental conditions. The coating resin shall not be applied to any part or surface where such application will interfere with the operations or performance of the equipment. Upon examination, the cured coating shall present a smooth, continuous surface. The coating shall pass from the mounted components to the circuit board with no break at their junction. Except for areas masked, all surfaces shall be covered with a coating of .001 inch minimum thickness. For exceptions to coating see 3.2.8.4.

3.2.8.3 Cured coating.- The cured coating shall meet the following requirements:

(a) Coating shall be free of visible dust particles or other foreign matter

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(b) It shall be smooth and uniform and have no globules or bare spots.

(c) It shall be substantially free of bubbles

(d) There shall be no blisters or cracks

(e) It shall be transparent and shall not affect marking legibility

(f) It shall be capable of being removed in local areas for replacement of components by a soldering iron

(g) When applied to a test board of 1/16 inch thickness, the coating shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius

3.2.8.4 Wire wound resistors. - One, two, three, and five watt wire wound resistors shall not be filleted or cemented to the board, nor shall they be coated.

3.2.9 Mounting requirements for stud mounted components. - The retaining nuts on stud mounted components shall be torqued as follows:

Transistors:

Thrd. Size	Torque Req't's Inch Lbs.
6-32	2-5
8-32	3-6
10-32	7-10
1/4-28	12-15
5/16-24	13-15
1/2-20	3-6

Diodes:

10-32	12-15
1/4-28	20-23
5/8-18	100-110

3.3 Wiring.

3.3.1 Wiring and harnesses. - All wiring shall be routed to provide reliability, to prevent malfunctioning of equipment by stray coupling, to provide accessibility for inspection and maintenance, and to minimize the possibility of damage. Wiring shall be routed in accordance with the applicable wiring diagram, to eliminate radio interference. Protective means, both physical and electrical, shall be employed to provide reliability and safety. Wires and cables attached to assemblies where relative movement occurs, such as at hinges or rotating pieces, shall be installed or protected in such a manner as to prevent deterioration of the wires and cable. Where cables or wires pass in close contact around any sharp corner of the equipment, such corners shall be covered and rounded with suitable cushioning material.

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3.3.2 Spot ties.- Groups of wires shall be securely bound by means of spot ties spaced at a distance 1/2 inch apart or twice the diameter of the wire group, whichever is greater. Spot ties shall be secured by a clove hitch and a square knot and have a dyed varnish placed on the knot. The cord diameter or ribbon width shall be suitable for the wire group diameter being tied. The spot ties shall be made of nylon which meets the requirements of MIL-T-713, Type P, and shall be fungus proofed in accordance with MIL-T-3530.

3.3.3 Cable support.- Connections within the equipment shall be suitably supported in such a manner as to prevent breakage and eliminate changes in frequency or output. The use of a flexible and a solid conductor in a continuous run without a support at their junction is not acceptable. Wires and cables shall be supported at intervals of not more than 12 inches. Wires shall not be tied or fastened together in conduit or insulating tubing. Moisture-absorbent type materials shall not be used as a fill for clamps. Cables fastened to two members not mechanically rigid shall be securely clamped to each member with sufficient slack so that there will be no strain or chafing of the cable.

3.3.4 Service loops.- Unless otherwise specified in the applicable wire list, wires and cables shall be provided with service loops. The service loop shall provide excess wire sufficient to permit two re-terminations. The re-termination factor shall be determined by the size and type of the terminal and the stripping length necessary for efficient re-termination.

3.3.5 Wire termination.- When soldering wires to terminals or solder pots, a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build-up. Burned, frayed, or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without disarranging, cutting, or otherwise damaging any strand of the conductor. Heat shall not be used to remove outer jacket from shielded wire. Only mechanical methods of removal are permitted.

EXCEPTION: The hot wire stripping method may be used where the wire or cable primary insulation and jacket are made of Teflon (TFE) or equal.

3.3.5.1 Shield termination.- After termination the shielded wire shall exhibit, when measured at 500 vdc, an insulation resistance no less then the insulation resistance of a piece of the same wire without termination. Exposed shielding shall not be allowed. Any exposed shielding shall be protected with insulation sleeving held in place with lacing cord or other restraining means. The insulation sleeving shall be in accordance with Specification MIL-I-23053. The maximum length of the shield ground wire as measured from the shield termination to the ground point shall not exceed 1.5 inches unless otherwise specified.

3.3.6 Connector wiring (miniature).- Care shall be exercised in wiring, soldering, and assemblying miniature and sub-miniature connectors so that the floating action of the connector inserts will not be impaired. Precaution shall be exercised in handling the connector before and after installation, so that no contamination, i.e., solder, flux, oil, or dust, etc. shall be deposited on the contact surfaces. The solder area shall be restricted to the

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contact terminal and neither solder nor flux shall be permitted to penetrate into the connector body. Pin-type connector contacts shall be protected from damage during assembly operations. There shall be no damage to the plated finish on the pins which is sufficient to expose the base metal.

3.3.7 Insulating sleeves.- Snug fitting insulating sleeving shall be placed on all wires terminating at connectors other than printed circuit and potted connectors. The sleeving shall be long enough to cover the terminal and at least 3/16 inch of the wire which is attached to it. The sleeving shall be shrunk on or tied with lacing cord with no more than 5 sleeves retained by any one tie. If the entire bundle is to be tied, the tie shall be placed beyond the end of the sleeving to prevent slippage in excess of 1/32 inch. Insulating sleeving shall be in accordance with Specification MIL-I-23053.

3.3.8 Connector damage.- In the event of any damage to or breakage of the connector body, floating bushing or threaded insert, the connector shall be removed and replaced with a new connector. Deformation of any contacts of a pin and socket type connector shall be cause for replacement of the connector. Connector contacts shall at no time be altered, depressed by probing, or otherwise readjusted. Pushout of one or more of the contacts from the body of a pin and socket type connector, as a result of wiring or other processing shall be cause for replacement of the connector.

3.3.9 Grommets.- All unused grommet holes in AN, MS and Miniature Type "E" connectors shall be filled with polyvinyl rod of durometer hardness approximately A90. When installed, the end of the polyvinyl rod shall extend 1/8 to 1/4 inch beyond the end of the grommet. All rods shall be trimmed to approximately equal length on the finished connectors.

3.3.10 Connector potting.- The potting will produce a "synthetic rubber" compound having suitable dielectric properties for moisture-proofing of the area to which applied. The cured potting compound shall be free from cracks and shall have a smooth, clean surface. Surface holes or bubbles smaller than 1/32 inch in diameter shall not be cause for rejection.

3.3.11 Potting requirements.- The depth of potting compound over the end of the wire insulation shall extend to a minimum of 3/16 inch. The compound shall adhere firmly to the connector backshell and shall completely enclose the wires. The exterior surfaces of the connector shall be free from potting compound. If sponging of the compound has occurred, as indicated by the presence of numerous small holes or a porous texture, the connector shall be rejected. If the connector lubricant has been removed from the external threads, apply a thin coat of lubricant conforming to Specification MIL-G-23827.

3.3.12 Grounding.- All areas to be grounded shall be cleaned of dirt, grease, paint, or other foreign matter in order to insure permanency and minimum resistance of the grounding connection. Any remaining exposed area from which a protective finish has been removed shall be refinished. When protective coatings have been removed to meet the requirements of this specification, the exposed area which remains after assembly, shall be refinished within 24 hours after inspection. A transparent finish, in accordance with Specification MIL-L-6806,

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and with a light amber soluble dye, no darker than color 13695 of Standard FED-STD-595, shall be used.

3.3.13 Chassis cleaning.- Chassis, cases, and wired parts shall be thoroughly cleaned and shall be free from superfluous particles of solder and all other foreign material. Cleaning processes shall have no visible or latent deleterious effect on the equipment. All traces of corrosive compounds shall be removed. Compressed air shall not be used for cleaning Vibration shall be used to clean foreign material out of the chassis after completion of the assembly operations.

3.3.13.1 Continuity assurance.- Subsequent to the vibration the computer's chassis (Drawing F10526102) shall exhibit electrical continuity in accordance with the requirements of the applicable wire lists.

3.3.14 Soldering and tinning.

3.3.14.1 Tinning.- All surfaces to be solder joined shall be tinned with solder prior to joining.

3.3.14.2 Soldering.- Soldering shall be so executed that both a positive electrical and machanical connection is assured. The solder must wet all surfaces with which it is in contact. The surface of the solder must be bright with no evidence of discontinuities. Solder shall conform with Specification QQ-S-571 using SN60 or SN63.

3.4 Threaded fasteners.

3.4.1 Screws and bolts.- All screw-type fasteners shall be tight. (The word "tight" shall mean that the screw cannot be tightened further without possible damage to the screw threads or material.)

3.4.2 Thread projection.- Screws, secured by nuts, or other retaining devices which permit projection beyond the retaining device, shall be of such length to permit a minimum projection of 1-1/2 threads and a maximum projection of 1/8 inch plus 1-1/2 threads for screws over 1 inch in length. These requirements shall not be construed as precluding the use of screws assembled in blind tapped holes in castings, or spacers, or where the design restrictions require that the screw length be such as to require that the threaded portion of the screw be flush with the retaining device. Screws and bolts in tapped parts other than nuts shall enter and engage for a distance at least equal to the diameter of the screw or bolt. Taps shall not be used for the cleaning of internally threaded parts.

3.4.3 Metallic inserts.- When screw threads engage tapped holes in nonmetals, an approved retaining compound shall be applied to such threads, or the tapped holes shall be provided with metallic inserts. Metallic inserts shall be staked in position, or secured by knurling or other method, to preclude their rotation. Inserts shall be of such design that tightening of the screws or studs, which they are intended to secure, will not result in the loosening or movement of the inserts. Metallic inserts in nonmetals shall be provided for those applications where additional strength is necessary, or where the screw will be removed and replaced frequently in service.

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3.4.4 Locking of threaded adjustments. - Shafts and adjusting screws of variable resistors, or other adjusting devices which are set at the factory during functional tests, shall be secured to maintain adjustment under specified service conditions.

3.4.5 Lock-wiring. - Lock-wire shall be used only when specified on the drawings. Parts shall be lock-wired in such a manner that the lock-wire shall be put in tension when the part tends to loosen. The pigtail end of the wire shall have a minimum of three and a maximum of six twists and shall be bent back or under (in a direction to increase the tension) to prevent it from becoming a snag. Lock-wire shall be installed only one time; destroy if removed for any reason. Avoid sharp bends, kinks, and nicking in lock-wire.

3.4.6 Dyed washers. - Washers which have been color coded for identification purposes shall be cleaned prior to installation on terminal boards, bus bars, etc.

3.5 Marking.

3.5.1 Impression stamping. - Impression stamps shall not crack or distort parts. Whenever possible, parts shall be impression stamped before application of protective finishes; however, impression stamps applied over cadmium or zinc plate and anodize are acceptable. Impression stamping after anodize shall be treated with iridite. The characters of the impression stamp shall have rounded edges. Impression depth shall be sufficient to insure legibility but not more than 0.005 inch. Unless otherwise specified on the detail drawing, all characters shall be standard block type or gothic.

3.5.2 Silk screen and rubber stamp markings. - The markings shall be free from distortion with clear and sharp edges. Identifications shall be marked in a color which provides a satisfactory contrast with the background (such as white on black or black on white). All rubber stamped markings shall be protected with a clear varnish. Unless otherwise specified, all markings shall be standard block type or gothic. Inks used for marking purposes shall be non-conductive and non-corrosive.

3.5.2.1 Marking durability.

(a) Abrasion. - Silk screen and rubber stamp markings shall show no evidence of wear or loss of legibility after being subjected to a minimum of 150 cycles of a GS-17 abrasive wheel with a pressure of 1000 grams.

(b) Weathering. - Subsequent to exposure to an accelerated weathering test as specified in ASTM Standard 750-55T for a period of 200 hours, there shall be no evidence of loss of legibility of discoloration in the silk screen or rubber stamp markings.

3.5.3 Etching and engraving. - Etching and engraving on panels shall be to a depth of either 0.011 $\pm$ 0.001 inch or 50 percent of the thickness of the metal, whichever is less. The etching or engraving shall be of such quality to produce

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clearly defined and sharply outlined characters. Panels used for any test noted herein except the gloss test shall be discarded after completion of the test. The backgrounds of engraved or etched panels shall be painted or dyed anodic film. Etched or engraved characters in the panel shall be filled using baking enamel. (see 6.2b and c).

3.5.4 Serialization labels.- The technique of application shall not damage the components or affect the leads by intolerable bending. This method of labeling shall be used only when specified on the drawing. Labels, when removed from the adhesive-protecting backing and in the course of application to the components, shall be carefully manipulated to avoid contaminating the adhesive surface in any manner which would impair the bond strength. The applied labels shall be legible and properly positioned.

3.5.5 Marking ink for non-metals.- Marking ink shall meet the following requirements:

(a) Shall be electrically non-conductive and non-corrosive

(b) All characters shall be opaque, legible, and clearly defined

(c) Black or white colors shall be used unless otherwise specified.

3.5.6 Application of decals.- The area to which the decal is to be applied shall be cleaned with naptha, Specification TT-N-95, and smoothed prior to application. Immediately prior to application, the area to which the decal is to be applied shall have a thin coat of varnish, Specification MIL-V-6894, applied to it and allowed to become tacky (10-20 minutes). The decal shall then be positioned. Excess water and entrapped air shall be removed by use of a soft roller. Any lack of adhesion shall be cause for replacement of the decal.

3.6 Finishes.- Finishes shall be free from dried overspray, runs, sags, blisters, orange peel effect, and other imperfections. Adhesion of finishes shall be determined on sufficient parts or assemblies to ensure that adhesion requirements are being met (see 6.2d).

3.6.1 Finish adhesion test.- The finish on all panels shall exhibit no more than ten percent loss in adhesion after 24 hours immersion at 77° to 80°F in water and Type III test fluid, Specification TT-S-735.

3.6.2 Finish gloss.- Finish gloss shall be measured in accordance with ASTM Standard D523-53T at a 60° angle with a glossmeter calibrated at a gloss standard value of 23. A total of five readings shall be taken one each at five different places on a completed panel. Each reading shall not exhibit a value greater than 10.

3.6.3 Finish durability.- Subsequent to exposure to an accelerated weathering test as specified in ASTM Standard D750-55T for a period of 200 hours there shall be no evidence of cracking, peeling or other film effects in the finish.

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3.7 Handling.

3.7.1 Mechanical and machine parts. Mechanical and machine parts having external finished surfaces of 63-500 micro-inches surface roughness shall be protected during in-process handling and storage by the use of totepans, containers, racks, pinboards, or handling baskets. Machined parts having external finished surfaces of 16-32 micro-inches surface roughness shall be protected during in-process handling by the use of individual containers, or separated from each other by using compartmented dividers in totepans, containers, racks, or handling baskets. Machined parts having external finished surfaces less than 16 micro-inches surface roughness shall be protected individually during in-process handling and storage by individually packaging in standard containers, bags, or wraps.

3.7.2 Cleaning of machined parts. - Machined parts shall be cleaned in accordance with the applicable drawings. Where such instructions do not exist, cleaning shall be selected to insure against any degradation of the part, or material.

CAUTION: Parts subject to fingerprint corrosion must not be handled with bare hands after cleaning.

3.7.3 Corrosion protection of machined parts. - Machined parts, subassemblies, and assemblies shall be protected against corrosion immediately following cleaning. Packaging materials in direct contact with the preservative shall conform to Grade A, Type I or II, of Specification MIL-B-121.

3.7.4 Electronic parts. - Small electronic parts such as relays, coils, terminals, meters, plugs, connectors, receptacles, etc., shall employ the vendor's package for physical protection wherever possible.

3.7.5 Connectors. - Plastic protective dust caps shall be placed on all connectors except when the connectors are being installed or used with a mating connector. The dust covers shall remain installed on the electrical connector throughout manufacturing, in-process handling, and during final shipment. When assembly or functional test operations require removal of the protective cover, it shall be removed and immediately replaced after performing the required operations when called out on applicable drawings. Metal type protective covers attached to the connectors or assembly may be used in lieu of plastic dust caps. Pressure sensitive tapes shall not be used to provide protection on connectors except in soldering and potting operations.

3.7.6 Assemblies. - Subassemblies and assemblies shall be physically protected during in-process handling and storage by the use of standard containers, totepans, racks, handling baskets, or other suitable devices. Critical electronic subassemblies shall be kept separated from each other by using individual containers, or compartmented toteboxes, or handling containers.

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3.7.7 Bearings.- Precision bearings shall be maintained in vendor's original package up to the point of installation into a sub-assembly or assembly. Sub-assemblies and assemblies containing precision bearings shall be maintained in the highest degree of cleanliness by the use of plastic shrouds, wraps, or covers to prevent contamination. When assembly or functional test operations require removal of the shroud, wrap, or cover, it shall be removed and replaced after performing the required operation.

3.7.8 Rubber materials.- Rubber materials shall be kept clean and dry at all times. If removed from original package and not installed, they shall be immediately returned to the package to retain identification of the part and cure date and to reduce age deterioration.

3.7.9 Matching surfaces.- Parts and sub-assemblies having matching surfaces to effect seals, with or without the use of "O" rings or gaskets, shall be provided physical protection against scratching or denting.

3.8 Transformers.- Materials, fabrication and workmanship shall be in accordance with the applicable drawings and the requirements of MIL-T-27.

3.9 Solvent vapor degreasing.- Oily and greasy parts may be cleaned in a solvent vapor degreaser when it is necessary or desirable that they be clean for easier handling, for easier working or for application of protective films during fabrication. Assemblies with non-metallic portions and parts such as oil impregnated sintered bearings, wire braid covered bellows with dri-film lube, etc., which might be damaged by vapor degreasing shall be degreased only with specific approval. Tetrachloroethylene (O-T-236) is intended specifically for use in the "vapor only" type degreasers. Trichloroethylene (MIL-T-7003) may be used in all applications.

3.9.1 Holding fixtures.- All holding fixtures such as racks and baskets and their associated parts such as chains, hooks, etc., shall be of metal.

3.9.2 Control of p^H and temperature.- The degreaser solvent shall be maintained at a p^H of not lower than 7.0. Solvent shall be maintained at a temperature sufficient to maintain a vapor phase, but shall not be allowed to exceed 195°F.

3.9.3 Extra-clean requirements.- If the drawing calls for vapor degreasing to "Extra-Clean" requirements, the following shall be applied in addition to other requirements. A jet of atomized water shall be sprayed on the inspected surface with acceptable cleanliness being indicated by coalesence of the water droplets to a continuous film on the area tested. Only distilled water shall be used for this test.

3.10 Bearings.

3.10.1 Precautions during bearing assembly.- Operations described for bearings shall be carried out in a dust-free area. Bearing washing and drying areas shall be provided with forced air evacuation so that solvent fumes will be exhausted as soon as formed. If dimensional measurements are required, they shall be made after bearings have stabilized at room temperature.

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3.10.2 Materials.- Lubricating oils, finger print removers, preservatives and solvents shall be stored in glass bottles having ground glass stoppers. Materials shall be passed through filters capable of removing particles 2 microns in size or more, and which discharge directly into the storage bottles. Solid lubricants shall be stored in containers having removable air-tight lids.

3.10.3 Demagnetization.- Bearings shall only be brought into the zone of influence (within 4 feet) of the demagnetizer during actual demagnetization operations. Bearings which are inadvertently brought into the zone of influence shall be checked by the Demagnetization Checker and, where necessary, again be demagnetized. Bearings shall be demagnetized before removal from the container in which received. Racks, trays, and other handling equipment shall be made of glass, solvent resistant plastic, or non-magnetizable metals. The bare hands or skin shall not be permitted to contact bearings or their components at any time. If dimensional measurements are required, they shall be made after bearings have stabilized at room temperature.

3.11 Environment.

3.11.1 Dust controlled area.- The dust particle size shall be not greater than 5 microns nor shall there be more than 200 particles within the range of 0.5 to 5.0 microns per cubic foot.

3.11.2 Temperature-controlled area.- The temperature in a temperature controlled area shall be such that it does not fluctuate more than $\pm$ 5 degrees Fahrenheit from the mean level for which the system is adjusted.

3.11.3 Maximum humidity - controlled area.- The relative humidity shall be maintained below 50 percent.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.1.1 Screening.- The requirements and tests listed below shall be inspected on a 100 percent basis at the standard ambient temperature (60° to 90°F).

100 percent inspection

Characteristics	Requirement	Method of Inspection
(a) Connector damage	3.3.8	Visual & Tactile
(b) Continuity assurance & Chassis cleaning	3.3.13, 3.3.13.1	Std. Meas. Equipment

4.2.2 Examination.- Inspection by examination to determine compliance with Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Tables I to VIII inclusive. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - Etched Circuit Assembly

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Laminated reinforced plastic sheet	3.2.1(a)	MIL-P-18177, Visual (Vis.)
102. Electrolytic copper (weight, Thickness and purity)	3.2.1(b)	Std. Meas. Equip. (SME)
103. Copper clad laminate sheets	3.2.1(c)	SME
104. Copper clad materials	3.2.1(d)	MIL-P-13949, Vis.
105. Installation of hardware	3.2.2(a) to j incl.	Vis. & SME
106. Cleaning of plating and etching	3.2.3.1	Visual
107. (Deleted)		
108. Plating (gold)	3.2.3.2(a)(c)	MIL-G-45204, Vis., SME
109. Plating (gold-nickel)	3.2.3.2(b)(c)	QQ-N-290, Vis., SME
110. Double and singled sided boards (feed through holes)	3.2.3.3.1 & 3.2.3.3.2	Visual & SME
111. Plated etched circuit boards	3.2.3.5(a) to (g)incl.	Visual
112. Registration	3.2.3.5.1	SME

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TABLE I - Etched Circuit Assembly (Continued)

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS		REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE			
MAJOR: AQL 1.0% Defective			
113.	Mounting of Components	3.2.4	Visual
114.	Spacing of parts	3.2.4.2	Visual & SME
115.	Pressure points	3.2.4.2.1	Visual
116.	Parts with welded leads	3.2.4.3	Visual
117.	Clinching of leads	3.2.5.3	Visual
118.	Transistor and diode leads	3.2.5.4	Visual & SME
119.	Securing leads	3.2.5.5	Visual
120.	Soldering	3.2.6.1	MIL-S-6872, Visual
121.	Tinning	3.2.6.2	Visual & SME
122.	Flux	3.2.6.3	MIL-F-14256, Visual
123.	Terminals	3.2.6.4	Visual
124.	Plated through connections	3.2.6.5	Visual
125.	Heat shunts	3.2.6.7	Visual
126.	Filleting or cementing components	3.2.8.1	Visual & SME
127.	Coating	3.2.8.2	Visual & SME
128.	Curing coating	3.2.8.3(a) to (g) incl.	Visual & SME
129.	Wire wound resistors	3.2.8.4	Visual
130.	Stud mounted components (Torque)	3.2.9	Visual & SME
MINOR: AQL 4.0% Defective			
201.	Conductor pattern	3.2.3.5	Visual & SME
202.	Symbols and masking	3.2.4.1	Visual
203.	Uninsulated parts	3.2.4.2.2	Visual & SME
204.	Large or heavy parts	3.2.4.4	Visual
205.	Mounting supports	3.2.4.5	Visual
206.	Metal case components	3.2.4.6	Visual
207.	Lead bends	3.2.5	Visual
208.	Lead length and bend radius	3.2.5.1, 3.2.5.2	Visual & SME
209.	Component leads	3.2.6.6	Visual & SME
210.	Cleaning	3.2.7	Visual
211.	Drying	3.2.7.1	Visual & SME

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TABLE II - Wiring

Use Inspection Level II of Table I With
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0%Defective		
131. Wiring & harnesses	3.3.1	Visual (Vis.)
132. Spot ties	3.3.2	MIL-T-713, Visual
133. Wire termination	3.3.5	Visual & SME
134. Shield termination	3.3.5.1	MIL-I-631, STD. Meas. Equip. (SME), Visual
135. Shield termination (insulation resistance)	3.3.5.1	MIL-I-23053, SME
136. Connector wiring (miniature)	3.3.6	Visual
137. Soldering & tinning	3.3.14	QQ-S-571, Visual
MINOR: AQL 4.0% Defective		
212. Cable support	3.3.3	Visual
213. Service loops	3.3.4	Visual
214. Insulating sleeves	3.3.7	Visual & SME, MIL-I-23053
215. Grommets	3.3.9	Visual & SME
216. Connector potting	3.3.10	Visual & SME
217. Potting requirements	3.3.11	Visual & SME
218. Grounding	3.3.12	FED-STD-595, Visual

TABLE III - Threaded Fasteners

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
138. Screws and bolts	3.4.1	Tactile
139. Metallic inserts	3.4.3	Visual & Tactile
140. Locking of threaded adjustments	3.4.4	Tactile
141. Lock wiring	3.4.5	Visual & Tactile
MINOR: AQL 4.0% Defective		
219. Threaded projection	3.4.2	Visual & SME
220. Dyed washers	3.4.6	Visual

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TABLE IV - Marking

Use Inspection Level II of Table I With
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
142. Impression stamping	3.5.1	Visual
143. Silk screen & rubber stamp marking	3.5.2	Visual
144. Application of decals	3.5.6	Visual & Tactile
MINOR: AQL 4.0% Defective		
221. Serialization labels	3.5.4	Visual & Tactile
222. Marking ink for non-metals	3.5.5(b) & (c)	Visual

Use Special Inspection Level S-4 of Table I
With Sampling Plan Table II-A of MIL-STD-105
for the following:

CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
145. Abrasion	3.5.2.1(a)	SME
146. Weathering	3.5.2.1(b)	ASTM-750-55T, Visual
147. Etching and engraving	3.5.3	Visual & SME
MINOR: NONE		

TABLE V - Finishes

Use Special Inspection Level S-4 of Table I With
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
148. Finish adhesion test	3.6.1	TT-S-735
149. Finish gloss	3.6.2	ASTM STD. D523-53T
150. Finish durability	3.6.3	ASTM STD. D750-55T
MINOR: NONE		

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TABLE VI - Handling

Use Special Inspection Level S-4 of Table I
With Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
151. Mechanical and machine parts	3.7.1	Visual
152. Cleaning of machined parts	3.7.2	Visual
153. Corrosive protection of machined parts	3.7.3	MIL-B-121, Visual
154. Electronic parts (packaging)	3.7.4	Visual
155. Connectors (protection)	3.7.5	Visual
156. Assemblies (protection)	3.7.6	Visual
157. Bearings (protection)	3.7.7	Visual
158. Rubber materials (protection)	3.7.8	Visual
159. Matching surfaces (protection)	3.7.9	Visual

TABLE VII - Bearings

Use Inspection Level II of Table I With
Sampling Plan Table II-A of MIL-STD-105

CLASSIFICATION OF DEFECTS	REQUIREMENT	METHOD OF INSPECTION
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
160. Precautions during bearing assy.	3.10.1	Visual
161. Materials (lubricant, cleaners, etc.)	3.10.2	Visual
162. Demagnetization	3.10.3	Visual & SME
MINOR: NONE		

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4.2.3 Certification.- Certification will be accepted for the requirements in Table VIII and shall include test data and results for all characteristics specified. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will require a new certification from the contractor.

TABLE VIII - Certification

CHARACTERISTIC		REQUIREMENT
301.	Circuit board material (copper)	3.2.1(b)
302.	Circuit board material (gold plating)	3.2.3.3(a)
303.	Circuit board material (gold-nickel plating)	3.2.3.3(b)
304.	Solder-Type SN60 or SN63 (QQ-S-571)	3.2.6.1
305.	Drying	3.2.7.1
306.	Coating (resin)	3.2.8.2
307.	Spot ties (fungus)	3.3.2
308.	Grommets (durometer hardness)	3.3.9
309.	Impression stamping (.005 depth)	3.5.1
310.	Marking ink for non-metals	3.5.5(a)
311.	Marking ink for decals (varnish)	3.5.6
312.	Transformers	3.8
313.	Solvent vapor degreasing	3.9
314.	Bearing assembly (dust free area)	3.10.1
315.	Environment	3.11

4.3 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.3.1 Contractor design.- The contractor shall provide inspection equipment compatible with the 4.3.1.1 and the following tolerance criterion for all standard measuring equipment specified in Tables I to VII inclusive. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.3.1.1 Test equipment.- In conjunction with 4.3.1, the following test equipment shall be utilized in the performance of the applicable tests:

ITEM NUMBER		DESCRIPTION
(a)	Holding fixtures (see 3.9.1)	To be constructed of metal
(b)	Control of P_H and temperature for solvent vapor degreasing (see 3.9.2)	Within ± 2%
(c)	Extra clean requirement (see 3.9.3)	Distilled water
(d)	Demagnization	Standard shop type
(e)	Oils - cleaning solvent finger print remover (see 3.10.2)	Pure within 2 micro
(f)	Dust control (see 3.11.1)	Filtered air-pure within 5 microns of dust, less then 200 particles of dust within 0.5 to 5.0 microns per cu.. ft.
(g)	Temperature control (see 3.11.2)	± 5 degrees
(h)	Maximum humidity controlled area (see 3.11.3)	Below 50%

4.4 Test methods and procedures.- The requirements of Section 3 shall be inspected in accordance with 4.2.1.1 and the respective Tables I to VII inclusive. Test methods and procedures are identified under "Method of Inspection" and shall be conducted utilizing Standard Measuring Equipment (SME) as specified. Other testing shall be in accordance with the related test methods and procedures outlined in Section 4 of the specified specifications, or the tests outlined in other documents specified in the tables. Certification shall be in accordance with Table VIII.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in the individual detailed specification.

6. NOTES

6.1 Intended use.- This specification is intended for use in connection with the manufacturing and inspection of the Gun Direction Computer: M18.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Background color of etched or engraved panels (see 3.5.3).
c. Whether background of etched or engraved panels shall be painted or dyed anodic film (see 3.5.3).
d. Number of parts or assemblies to insure adhesion qualities (see 3.6).
e. Unless otherwise specified, printed circuit terms and definitions shall conform to MIL-STD-429.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A215Y

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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DEPARTMENT OF THE ARMY

Frankford Arsenal
Philadelphia, Pa. 19137

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DEPARTMENT OF THE ARMY

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-5500
Philadelphia, Pa. 19137

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